Derivatives of Elementary Functions

Reference Table and Proofs

1. Derivative reference table

All trigonometric angles are measured in **radians**.

Function $f(x)$	**Derivative** $f'(x)$	**Domain or conditions**
Constant C	0	$x \in \mathbb{R}$
x^α	$\alpha x^{\alpha-1}$	For arbitrary real α, first take $x > 0$
e^x	e^x	$x \in \mathbb{R}$
a^x	$a^x \ln a$	$a > 0$, $a \neq 1$
$\ln x$	$\dfrac{1}{x}$	$x > 0$
$\log_a x$	$\dfrac{1}{x \ln a}$	$x > 0$, $a > 0$, $a \neq 1$
$\sin x$	$\cos x$	$x \in \mathbb{R}$
$\cos x$	$-\sin x$	$x \in \mathbb{R}$
$\tan x$	$\sec^2 x = \dfrac{1}{\cos^2 x}$	$\cos x \neq 0$
$\cot x$	$-\csc^2 x = -\dfrac{1}{\sin^2 x}$	$\sin x \neq 0$
$\sec x$	$\sec x \tan x$	$\cos x \neq 0$
$\csc x$	$-\csc x \cot x$	$\sin x \neq 0$

A note on powers. For positive integer powers, the formula holds for every real x. Negative integer powers require $x \neq 0$. For other powers, behavior at zero or at negative inputs must be checked for the particular function.

Differentiation finds a function's **instantaneous rate of change** at a point. Every derivative formula ultimately comes from the same definition:

$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.}$$

2. The four differentiation rules

Suppose $u = u(x)$ and $v = v(x)$ are differentiable, and a, b are constants:

$$\boxed{\begin{aligned}(au + bv)' &= au' + bv',\\ (uv)' &= u'v + uv',\\ \left(\frac{u}{v}\right)' &= \frac{u'v - uv'}{v^2} \qquad (v \neq 0),\\ [f(g(x))]' &= f'(g(x))\, g'(x).\end{aligned}}$$

The last rule is the **chain rule**: differentiate the outer function with respect to its input, then multiply by the derivative of the inner function. For example,

$$(\sin x^2)' = \cos(x^2) \cdot 2x.$$

Why these rules hold

Linearity follows directly from the sum and constant-multiple rules for limits.

Product rule. The key is to split the simultaneous changes in the two factors:

$$\begin{aligned}&u(x+h)v(x+h) - u(x)v(x)\\ &\quad = [u(x+h) - u(x)]v(x+h)\\ &\qquad + u(x)[v(x+h) - v(x)].\end{aligned}$$

Divide by h and let $h \to 0$. Since differentiability implies continuity,

$$(uv)' = u'v + uv'.$$

Chain rule. An equivalent formulation of differentiability gives a rigorous proof. Let $\Delta u = g(x+h) - g(x)$. Then

$$\Delta u = g'(x)h + o(h),$$

while the change in the outer function is

$$f(g(x) + \Delta u) - f(g(x)) = f'(g(x))\Delta u + o(\Delta u).$$

Here $o(h)$ denotes an error that is negligible compared with h as $h \to 0$. Substitute, divide by h, and take the limit:

$$[f(g(x))]' = f'(g(x))g'(x).$$

Quotient rule. First differentiate the reciprocal:

$$\begin{aligned}\left(\frac{1}{v}\right)' &= \lim_{h\to 0} \frac{v(x) - v(x+h)}{h\, v(x+h)v(x)}\\ &= -\frac{v'}{v^2}.\end{aligned}$$

Now write $u/v = u \cdot (1/v)$ and apply the product rule.

3. Constants and powers: proofs

3.1 Constants

A constant does not change, so

$$(C)' = \lim_{h\to 0} \frac{C - C}{h} = 0.$$

3.2 Positive integer powers

Use the binomial expansion:

$$(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

Thus,

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}.$$

As $h \to 0$, every term still containing h tends to zero, leaving

$$\boxed{(x^n)' = nx^{n-1}.}$$

The proof for arbitrary real powers follows after exponentials and logarithms.

4. Logarithms and exponentials: proofs

Start with the familiar limit definition

$$\boxed{e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.}$$

4.1 Natural logarithms

Fix $x > 0$. Set $n = x/h$, so $h = x/n$:

$$\begin{aligned}
(\ln x)' &= \lim_{h\to 0} \frac{\ln(x+h) - \ln x}{h} \\
&= \lim_{h\to 0} \frac{\ln(1 + h/x)}{h} \\
&= \frac{1}{x} \lim_{|n|\to\infty} n \ln\left(1 + \frac{1}{n}\right) \\
&= \frac{1}{x} \ln\left[\lim_{|n|\to\infty} \left(1 + \frac{1}{n}\right)^n\right] \\
&= \frac{1}{x} \ln e = \boxed{\frac{1}{x}}.
\end{aligned}$$

The same defining limit gives e as $n \to -\infty$, so this works from either side. The change-of-base formula gives

$$\log_a x = \frac{\ln x}{\ln a}, \qquad \boxed{(\log_a x)' = \frac{1}{x \ln a}}.$$

4.2 Natural exponentials

Let $t = e^h - 1$. As $h \to 0$, we have $t \to 0$ and $h = \ln(1+t)$. Therefore,

$$\lim_{h\to 0} \frac{e^h - 1}{h} = \lim_{t\to 0} \frac{t}{\ln(1+t)} = 1.$$

The last limit is the reciprocal of the derivative of $\ln x$ at $x = 1$. Substitute into the definition of the derivative:

$$\begin{aligned}(e^x)' &= \lim_{h\to 0} \frac{e^{x+h} - e^x}{h} \\ &= e^x \lim_{h\to 0} \frac{e^h - 1}{h} = \boxed{e^x}.\end{aligned}$$

Since $a^x = e^{x \ln a}$, the chain rule gives

$$\boxed{(a^x)' = a^x \ln a}.$$

4.3 Arbitrary real powers

For $x > 0$,

$$x^\alpha = e^{\alpha \ln x}.$$

Therefore,

$$(x^\alpha)' = e^{\alpha \ln x} \cdot \frac{\alpha}{x} = \boxed{\alpha x^{\alpha-1}}.$$

Here α must be a **constant**. For example, in x^x the exponent also varies, so the power rule cannot be applied directly.

5. Trigonometric functions: proofs

First establish two limits:

$$\boxed{\lim_{h\to 0}\frac{\sin h}{h}=1}, \qquad \boxed{\lim_{h\to 0}\frac{\cos h-1}{h}=0}.$$

For the first, compare areas in the unit circle. When $0<h<\pi/2$,

$$\sin h < h < \tan h \quad \Longrightarrow \quad \cos h < \frac{\sin h}{h} < 1.$$

The squeeze theorem gives a right-hand limit of 1. Since $\sin h/h$ is even, the left-hand limit is also 1.

For the second limit, multiply by $1+\cos h$ on the top and bottom:

$$\frac{\cos h-1}{h}=-\frac{\sin^2 h}{h(1+\cos h)}=-\frac{\sin h}{h}\cdot\frac{\sin h}{1+\cos h}\longrightarrow -1\cdot\frac{0}{1+1}=0.$$

Here we just use $\sin h\to 0$ and $\cos h\to 1$ as $h\to 0$.

5.1 Sine and cosine

Use the angle-addition formula and the same algebraic rearrangement:

$$\begin{aligned}(\sin x)' &= \lim_{h\to 0}\frac{\sin x\cos h+\cos x\sin h-\sin x}{h}\\ &= \lim_{h\to 0}\frac{\sin h}{h}\left(\cos x-\sin x\frac{\sin h}{1+\cos h}\right)\\ &= 1\left(\cos x-\sin x\frac{0}{1+1}\right)=\boxed{\cos x}.\end{aligned}$$

Similarly,

$$\begin{aligned}(\cos x)' &= \lim_{h\to 0}\frac{\cos x\cos h-\sin x\sin h-\cos x}{h}\\ &= \lim_{h\to 0}\frac{\sin h}{h}\left(-\sin x-\cos x\frac{\sin h}{1+\cos h}\right)\\ &= 1\left(-\sin x-\cos x\frac{0}{1+1}\right)=\boxed{-\sin x}.\end{aligned}$$

5.2 The other four trigonometric functions

Write each as a ratio or reciprocal of sine and cosine, then differentiate:

$$(\tan x)'=\left(\frac{\sin x}{\cos x}\right)'=\frac{\cos^2 x+\sin^2 x}{\cos^2 x}=\boxed{\sec^2 x},$$

$$(\cot x)'=\left(\frac{\cos x}{\sin x}\right)'=\frac{-\sin^2 x-\cos^2 x}{\sin^2 x}=\boxed{-\csc^2 x},$$

$$(\sec x)'=\left(\frac{1}{\cos x}\right)'=\frac{\sin x}{\cos^2 x}=\boxed{\sec x\tan x},$$

$$(\csc x)'=\left(\frac{1}{\sin x}\right)'=-\frac{\cos x}{\sin^2 x}=\boxed{-\csc x\cot x}.$$

6. What is easiest to overlook in practice?

When x in a formula is replaced by a function, multiply by the derivative of that inner function.

For example,

$$[(3x+1)^5]' = 5(3x+1)^4 \cdot 3,$$

$$[\ln(1+x^2)]' = \frac{2x}{1+x^2},$$

$$[e^{-x^2}]' = -2xe^{-x^2}.$$

Also distinguish these three similar-looking functions:

$$\boxed{\begin{aligned} &(x^a)' = ax^{a-1}, \\ &(a^x)' = a^x \ln a, \\ &(x^x)' = x^x(\ln x + 1) \qquad (x > 0). \end{aligned}}$$

For the last formula, write $x^x = e^{x \ln x}$ and differentiate.

First identify whether the base, the exponent, or both are changing; then choose the formula.

7. Practice Exercises

Source: Joel Hass, Christopher Heil, and Maurice D. Weir, *Thomas' Calculus*, 14th edition, Pearson, 2018. ISBN 978-0-13-443898-6. References below use the book's printed page numbers.

Find the first derivative of each function with respect to its independent variable. Each answer follows its question; no working is included.

1. Source: Section 3.3, Exercise 17, p. 121.

$y = \dfrac{2x+5}{3x-2}$ $\left[\frac{dy}{dx} = -\frac{19}{(3x-2)^2}\right]$

2. Source: Section 3.3, Exercise 19, p. 121.

$g(x) = \dfrac{x^2-4}{x+0.5}$ $\left[g'(x) = \frac{x^2+x+4}{(x+0.5)^2}\right]$

3. Source: Section 3.5, Exercise 3, p. 138.

$y = x^2 \cos x$ $\left[\frac{dy}{dx} = 2x\cos x - x^2 \sin x\right]$

4. Source: Section 3.5, Exercise 7, p. 138.

$f(x) = \sin x \tan x$ $\left[f'(x) = \cos x \tan x + \sin x \sec^2 x\right]$

5. Source: Section 3.5, Exercise 12, p. 138.

$y = \dfrac{\cos x}{1+\sin x}$ $\left[\frac{dy}{dx} = -\frac{1}{1+\sin x}\right]$

6. Source: Section 3.6, Exercise 19, p. 145.
$p = \sqrt{3-t}$ $\left[\frac{dp}{dt} = -\frac{1}{2\sqrt{3-t}}\right]$

7. Source: Section 3.6, Exercise 29, p. 145.
$y = (4x+3)^4(x+1)^{-3}$ $\left[\frac{dy}{dx} = \frac{(4x+3)^3(4x+7)}{(x+1)^4}\right]$

8. Source: Section 3.6, Exercise 34, p. 145.
$g(x) = \dfrac{\tan 3x}{(x+7)^4}$ $\left[g'(x) = \frac{3(x+7)\sec^2(3x) - 4\tan(3x)}{(x+7)^5}\right]$

9. Source: Section 7.2, Exercise 17, p. 385.
$y = t(\ln t)^2$ $\left[\frac{dy}{dt} = (\ln t)^2 + 2\ln t\right]$

10. Source: Section 7.2, Exercise 25, p. 385.
$y = \ln(\ln x)$ $\left[\frac{dy}{dx} = \frac{1}{x\ln x}\right]$

11. Source: Section 7.3, Exercise 75, p. 395.
$y = x^3 \log_{10} x \quad \left[\frac{dy}{dx} = 3x^2 \log_{10} x + \frac{x^2}{\ln 10}\right]$

12. Source: Section 7.3, Exercise 7, p. 394.
$y = e^{-5x} \quad \left[\frac{dy}{dx} = -5e^{-5x}\right]$

13. Source: Section 7.3, Exercise 13, p. 394.
$y = (x^2 - 2x + 2)e^x \quad \left[\frac{dy}{dx} = x^2 e^x\right]$

14. Source: Section 7.3, Exercise 17, p. 394.
$y = \cos\left(e^{-u^2}\right) \quad \left[\frac{dy}{du} = 2ue^{-u^2} \sin\left(e^{-u^2}\right)\right]$

15. Source: Section 7.3, Exercise 69, p. 395.
$y = 2^{\sin 3t} \quad \left[\frac{dy}{dt} = 3(\ln 2)\, 2^{\sin 3t} \cos 3t\right]$